

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Computer Science & Engineering

2. **Subject Name:** Computer Networks

L-T-P: 3-1-0

Credits: 4

3. **Pre-requisites:** Algorithms-I (CS21203)

4. **Syllabus and reference books:**

Syllabus:

Introduction to networks and layered architecture. Basics of data communication: transmission media and topology, data encoding techniques, error control, flow control, multiplexing techniques. Medium access control: CSMA, CSMA/CD. Local Area Networks and Ethernet. Basics of circuit switching and packet switching. Routing protocols: distance vector and link state routing. IPv4: addresses, subnet, routing table and its use, route aggregation, RIP and OSPF, fragmentation, ICMP, ARP. Basics of IPv6. TCP: properties, connection establishment and termination, data transfer, flow control, congestion control. UDP, DNS, DHCP, case study of application layer protocols. Internet architecture and routing: physical interconnection, autonomous systems, BGP. Basics of wireless networks.

Reference Books:

- 1) Data and Computer communication, 10th Edition, by William Stallings, Pearson Education, 2017
- 2) Data Communication and Networking, 5th Edition, by Behrouz A. Forouzan, McGraw Hill, 2017
- 3) Internetworking with TCP/IP, Volume 1, by Douglas Comer, Person, 2013
- 4) TCP/IP Protocol Suite, 4th Edition, by Behrouz Forouzan, McGraw Hill, 2017
- 5) Computer Networking: a Top-down Approach, 7th Edition, by James Kurose and Keith Ross, Pearson, 2017

5. **Lecture-wise break-up:**

Sl. No.	Topic	No. of lectures
1.	Introduction to networks and layering	1
2.	Basics of data communication	8
3.	Medium access control	2
4.	LAN and Ethernet	3
5.	Basics of circuit and packet switching	1
6.	Routing protocols	2
7.	IPv4 and IPv6	6

8.	TCP and UDP	6
9.	DNS and DHCP	2
10.	Application layer protocols	2
11.	Internet architecture and routing	3
12.	Basics of wireless networks	4
Total number of hours		40